

OriginTrace — Oil-Spill Attribution Evidence Package

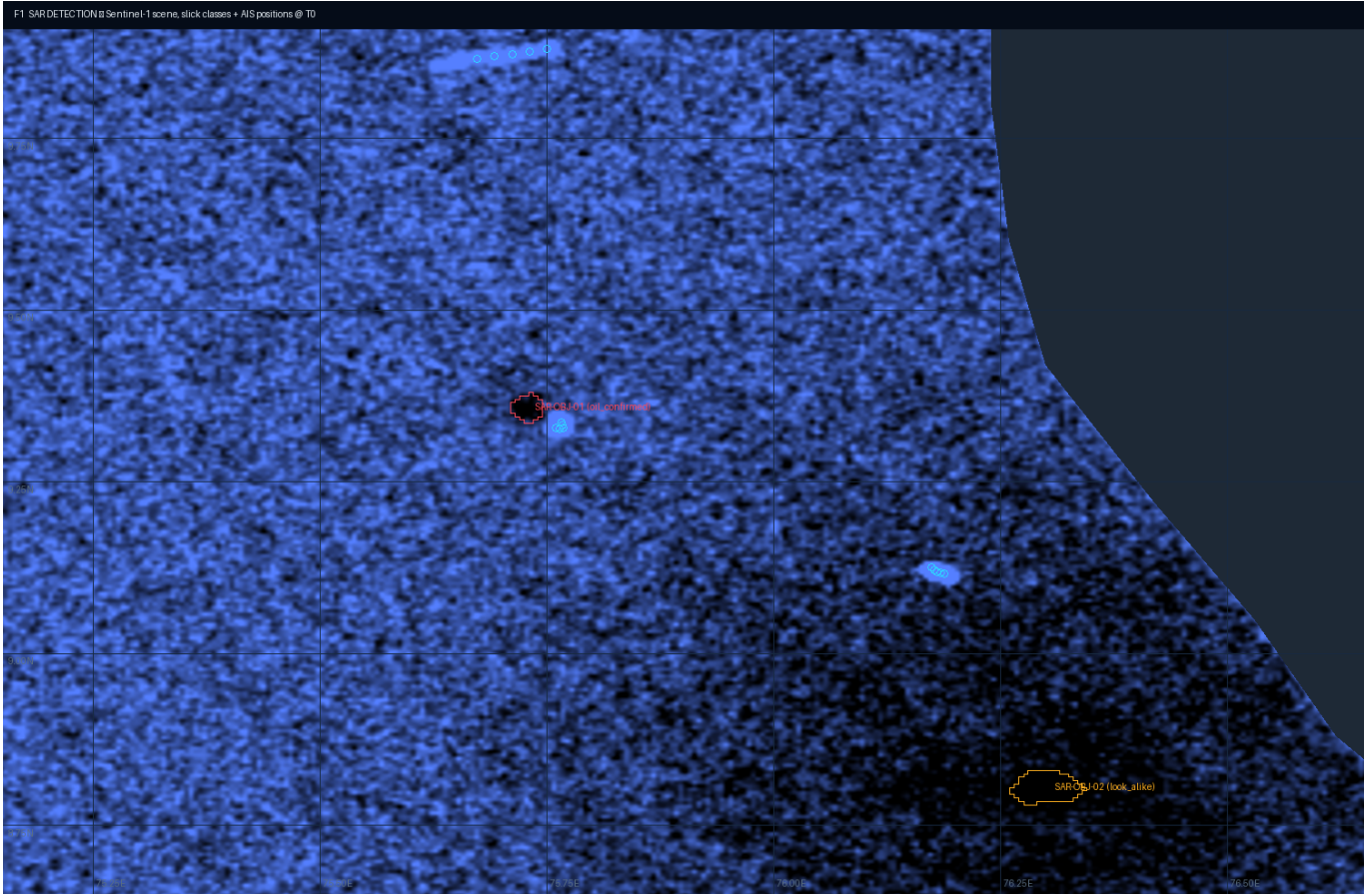
Case: **KERALA_2025_CASE01** | Scene: S1A_IW_GRDH_1SDV_20250612T063000_KERALA_SYNTH | Detection (UTC): **2025-06-12T06:30:00+00:00**
Generated: 2026-09-01 16:06Z | Prepared for: Indian Coast Guard (demo) | SIH-26143 Team Aryan

LEGAL TIER — UNCLOS Art. 220(3): TIP-AND-CUE. This package provides *clear grounds* to request information from the flagged vessels (name, port of registry, cargo log). It is **NOT** detention-grade evidence: Art. 220(5) physical inspection and Art. 220(6) detention require boarding and chemical fingerprinting beyond this system.

1 · Satellite Detection (Sentinel-1 SAR)

Object	Class	Area (km²)	Contrast (dB)	Wind (m/s)	Conf.
SAR-OBJ-01	oil_confirmed	11.0	2.48	7.2	0.77
SAR-OBJ-02	look_alike	29.8	1.86	1.7	0.35

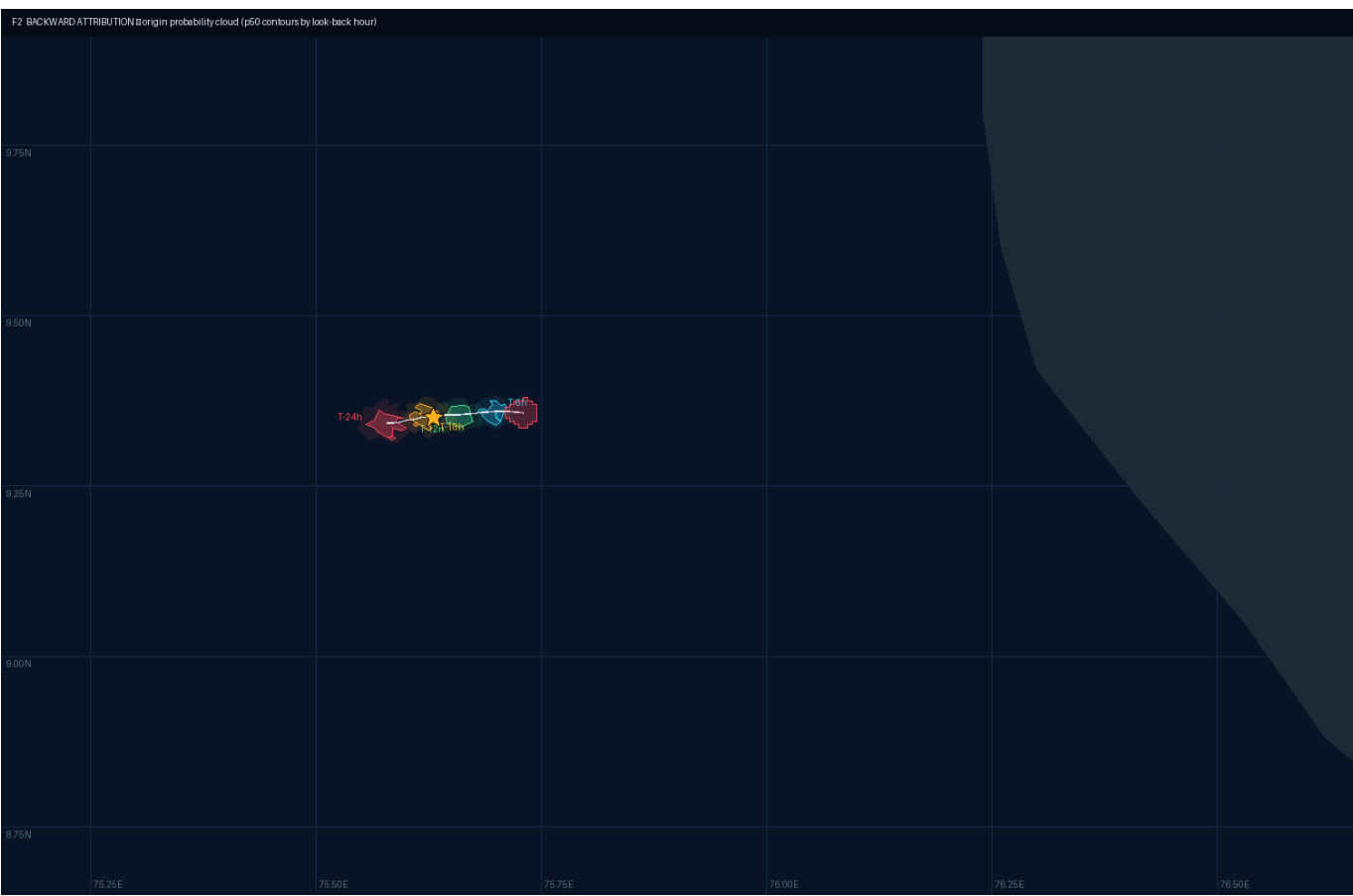
Detection is considered **reliable only in 3–12 m/s wind**; objects in low-wind are typed *look-alike* (see detectability mask on map layer).



2 · Backward Attribution (Origin Reconstruction)

350 Lagrangian particles advected **backward** 24 h from the detected slick (current + $0.03 \times U_{10}$ windage + diffusion, $dt=600$ s). Hourly KDE clouds $\rightarrow$ p10/p50/p90 probability-mass contours.

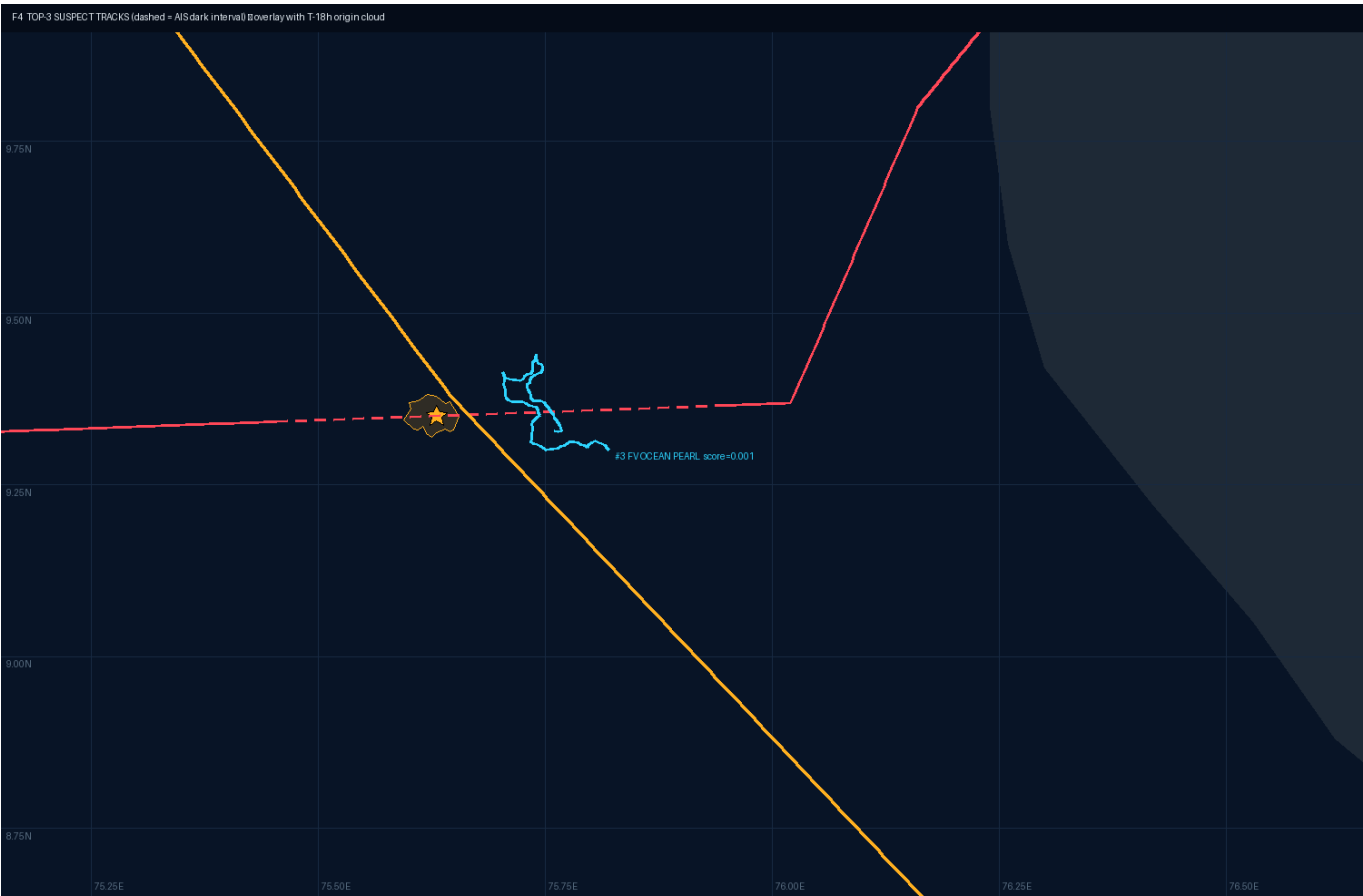
Estimated origin	Release window (UTC)	Engine
75.63E, 9.35N	2025-06-11T08:30Z -> 2025-06-11T16:30Z	OriginTrace Lagrangian Engine v1.0



3 · Vessel Attribution — Ranked Suspects

Model: LightGBM binary classifier (synthetic-calibrated) + TreeSHAP (pred_contrib) explanations. Release window scored: 2025-06-11T06:30Z → 2025-06-12T02:30Z. 20 vessels evaluated.

#	Vessel	MMSI	Verdict	Score	Leading factors
1	MT KAVERI STAR	419000001	PRIMARY SUSPECT	1.000	position coincides with the cloud at release-relevant hours; closest approach 0 km to the reconstructed release locus; vessel went dark for 6.2 h inside the release window
2	MV MALABAR TRADER	419000002	SUSPECT — CORROBORATED	0.790	position coincides with the cloud at release-relevant hours; best 3 h sub-window shows slow, straight-run steaming; closest approach 1 km to the reconstructed release locus
3	FV OCEAN PEARL	419000003	CLEARED	0.001	position coincides with the cloud at release-relevant hours; track intersects the reconstructed origin cloud at plausible times; closest approach 5 km to the reconstructed release locus



4 · Chain of Custody — SHA-256

Artifact	SHA-256
sar_scene (scene.tif)	8aa5c7904dcd2c22e6059b7ae6809ad870a4c8f5f07e9bebfac7cb8840f8e9bc
slick_polygons.geojson	7707b6f31f6ff16b6bcb8299e48d12138454b2be6b13d0aeb1991a56b839939d
backtrack_hourly.geojson	4bea61c458aa909905c4bdc1c176cf7faa28a9d7642a41e01978391b42a25134
suspects.json	4b61f5af7ec7fddf600df8a9dffa51efd3f9f797731a285ec5efcaf4c5ce31e
lgbm_ranker.txt	c50d51d5442eb2a30eaaaf40553a7e6b7ae30fc10384edd2270bbe4c5b8654c28

PACKAGE ID (hash of hashes)	4579a66e040ac261b611cda6a91e1fabf01489569a678af56a9c46471f0f4cb2
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5 · Electronic Evidence — Bharatiya Sakshya Adhiniyam 2023 §63

This package is structured for admissibility under **BSA 2023 §63** (electronic records): each artifact is identified by SHA-256 digest, generation timestamp, and processing chain. The PACKAGE ID above is the composite seal. Operational deployment would attach a qualified digital-signature certificate; this demo supplies the certificate *template* and hash chain only.

Synthetic demo case — generated scene/AIS/met-ocean stand-ins per the SIH-26143 build plan (Zenodo-fallback mode). Method chain is identical for operational Sentinel-1 / AIS feeds. OriginTrace is a decision-support system; all findings require corroboration per the UNCLOS enforcement ladder (220(5) inspection, 220(6) detention, MARPOL sampling).